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1. Solve the following $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 6y = 0$

Soln: Let $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} - 5 - 6y = 0$ be the LDE with CC.

Put $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$ and $D^3 = \frac{d^3}{dx^3}$

we get, $D^3y + 2D^2y - 5Dy - 6y = 0$

$(D^3 + 2D^2 - 5D - 6)y = 0$ —— ①

Consider the Auxiliary Equation (AE),

$D^3 + 2D^2 - 5D - 6 = 0$

$(D+1)(D-2)(D+3) = 0$

$\therefore D = -1, 2, -3$

∴ The roots are $-1, 2, -3$

∴ The complementary function is given as,

$C.F = C_1e^{-x} + C_2e^{2x} + C_3e^{-3x}$ [∵ X=0 the general solution is C.F only]

∴ The general solution of Equation ① is given as

$y = C_1e^{-x} + C_2e^{2x} + C_3e^{-3x}$

2. Solve $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$

Soln:- Let, $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$ be the given LDE with CC.

$(D^4y + 2D^2y + y) = 0$

$\therefore (D^4 + 2D^2 + 1)y = 0$ — ①

we have A.E as, $D^4 + 2D^2 + 1 = 0$

$(D^2+1)^2 = 0$

$(D^2+1)(D^2+1) = 0$

$(D+i)(D-i)(D+i)(D-i)\ D = \pm i, \pm i$

The roots of AE are $i, i, -i, -i$ [Here $\alpha=0, \beta=1$]

$C.F = e^{0x}[(C_1+C_2x)\cos x + (C_3+C_4x)\sin x]$

∴ The general solution is

$Y = [(C_1+C_2x)\cos x + (C_3+C_4x)\sin x]$

3. Solve $(D^4 - 2D^2 + 1)y = 0$

Soln: Let $(D^4 - 2D^2 + 1)y = 0$ be the given LDE with cc — ①

we have Auxiliary Equation as

$D^4 - 2D^2 + 1 = 0$

$(D^2-1)^2 = 0$

$(D^2-1)(D^2-1) = 0$

$[(D+1)(D-1)]^2 = 0$

$(D+1)^2 (D-1)^2 = 0$

∴ the roots of AE are $1, 1, -1, -1$

The general solution of Equation ① is given as,

$y = (C_1+C_2x)e^x + (C_3+C_4x)e^{-x}$

Problems on $X = e^{ax}$

1. Solve $(D^2+D-6)y = e^{4x}$

Soln:- Let $(D^2+D-6)y = e^{4x}$ — ① be the given LDE with cc

we have AE as $D^2+D-6 = 0$

$(D+3)(D-2) = 0$

$D = -3, 2$

The roots of AE are $-3, 2$ $\quad C.F = C_1e^{-3x} + C_2e^{2x}$ — ②

Consider, $PI = \frac{1}{D^2+D-6} \cdot e^{4x}$

$= \frac{1}{4^2+4-6} \cdot e^{4x} = \frac{1}{14} e^{4x}$

$PI = \frac{e^{4x}}{14}$

The general solution of Equation ① is given as,

$y = C.F + PI$

$y = C_1e^{-3x} + C_2e^{2x} + \frac{e^{4x}}{14}$

2. Solve $(D^2+D)y = e^{4x}$

Soln:- Let $(D^2+D)y = e^{4x}$ — (I) be the given LDE with CC

we have auxiliary Equation $D^2+D=0$

$D(D+1)=0$

$\therefore D=0,-1$

The roots of AE are $0, -1$

$\therefore CF = C_1 + C_2 e^{-x}$ — (II)

Now we have $P.I = \frac{1}{D^2+D} \cdot e^{4x} = \frac{1}{4^2+4} \cdot e^{4x}$

$= \frac{1}{20} \cdot e^{4x}$

$P.I = \frac{1}{20} \cdot e^{4x}$ — (III)

The general solution of Equation (I) is,

$y = C.F + P.I$

$y = C_1 + C_2 e^{-x} + \frac{1}{20} e^{4x}$

3. Solve $(D^2+D-6)y = \sinh 3x$

Soln: Let $(D^2+D-6)y = \sinh 3x$ — (I) be the given LDE with CC

we have Auxiliary Equation, $D^2+D-6=0$

$(D+3)(D-2)=0$

$\therefore D=-3,2$

The roots of AE are $-3, 2$.

$\therefore CF = C_1 e^{-3x} + C_2 e^{2x}$ — (II)

Now we have $P.I = \frac{1}{D^2+D-6}\left(\frac{e^{3x}-e^{-3x}}{2}\right)$

$\therefore P.I = \frac{1}{2}\left(\frac{e^{3x}}{6} - xe^{-3x}\right)$

$P.I = \frac{e^{3x}}{12} - \frac{xe^{-3x}}{2}$

$$y = C_1 e^{-3x} + C_2 e^{2x} + \frac{e^{3x}}{12} - \frac{xe^{-3x}}{2}$$

4. Solve $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = \cosh^3 x$

Soln:- Let $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = \cosh^3 x$ — ① be the given LDE with cc

Put $\frac{d^3y}{dx^3} = D^3$ we get $(D^3+4D)y = \frac{1}{4}(3\cosh x + \cosh 3x)$

we have AE as

$$D(D^2+4) = 0$$
$$D = 0, \pm 2i$$

∴ The roots of AE are $0, +2i, -2i$

Now C.F $= C_1 + C_2 \cos 2x + C_3 \sin 2x$

Now consider, P.I $= \frac{1}{4}\left[\frac{3}{D^3+4D}\cosh x + \frac{1}{D^3+4D}\cosh 3x\right]$

$f(1) = \frac{\cosh x}{5}$

$f(3) = \frac{\cosh 3x}{39}$

$\therefore P.I = \frac{1}{4}\left(\frac{3\cosh x}{5} + \frac{\cosh 3x}{39}\right)$ — ②

∴ The general solution of equation ① is

$$y = C.F + P.I$$
$$y = C_1 + C_2 \cos 2x + C_3 \sin 2x + \frac{3\cosh x}{20} + \frac{\cosh 3x}{156}$$

Problems on $X = \sin ax$ or $X = \cos ax$

1. Solve $(D^4 - a^4)y = \sin ax$

Soln:- Let $(D^4 - a^4)y = \sin ax$ be the given LDE with cc.

we have $D^4 - a^4 = 0$

$$(D^2 - a^2)(D^2 + a^2) = 0$$
$$(D-a)(D+a)(D-ai)(D+ai) = 0$$
$$D = a, -a, ai, -ai$$

The roots of AE are $a, -a, ai, -ai$

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Consider, $P.I = \frac{1}{(D^2+a^2)(D^2-a^2)} \cdot \sin ax$

$= \frac{1}{D^4-a^4} \cdot \sin ax$

$= \frac{x}{4D^3} \cdot \sin ax$

$= \frac{x}{4} \cdot \frac{1}{D^3} \sin ax$

$= \frac{x}{4} \cdot \frac{1}{D^2} \cdot \frac{1}{D} \sin ax = \frac{1}{4} \frac{1}{D^2} \int \sin ax \, dx$

$= \frac{x}{4} \frac{1}{D^2} \frac{\cos ax}{a}$

$P.I = \frac{x}{4} \cdot \frac{1}{D^2} \frac{\cos ax}{a} = \frac{x}{4a} \cdot \frac{1}{D} \cdot \frac{1}{D} \cos ax$

$= \frac{x}{4a} \frac{1}{D} \int \cos ax \, dx = \frac{x}{4a} \cdot \frac{1}{D} \left(\frac{-\sin ax}{a}\right)$

$P.I = \frac{-x}{4a^2} \cdot \frac{1}{D} \cdot \sin ax = \frac{-x}{4a^2} \cdot \frac{\cos ax}{a}$

$= \frac{-x}{4a^3} \cdot \cos ax$ — ③

The general solution of equation ① is

$y = C.F + P.I$

$y = C_1 e^{ax} + C_2 e^{-ax} + C_3 \cos ax + C_4 \sin ax - \frac{x}{4a^3} \cos ax$

2. Solve $(D^3+D^2+D+1)y = \sin^2 x$

Soln: Let $(D^3+D^2+D+1)y = \sin^2 x$ — ① be the given LDE with CC. we have AE $D^3+D^2+D+1=0$

$D^2(D+1)+1(D+1)=0$

$(D^2+1)(D+1)=0$

$D = -1, \pm i$

The roots of AE are $-1, i, -i$

$\therefore C.F = C_1 e^{-x} + C_2 \cos x + C_3 \sin x$ — (2)

Now we have,

$$P.I = \frac{1}{1+D+D^2+D^3} \cdot \sin^2 x = \frac{1}{(D^2+1)(D+1)} \left(\frac{1-\cos 2x}{2}\right)$$

$$P.I = \frac{1}{2} \frac{1}{(D^2+1)(D+1)} (1-\cos 2x)$$

$$= \frac{1}{2}\left[\frac{1}{(D^2+1)(D+1)} \cdot e^{2x} - \frac{1}{(D^2+1)(D+1)} \cdot \cos 2x\right]$$

$$= \frac{1}{2}\left[1 - \frac{1}{[-4+1][D+1]} \cdot \cos 2x\right]$$

$$= \frac{1}{2}\left[1 - \frac{1}{-3(D+1)} \cos 2x\right]$$

$$= \frac{1}{2}\left[1 + \frac{(D-1)}{3(D^2-1)} \cdot \cos 2x\right]$$

$$= \frac{1}{2}\left[1 - \frac{1}{3(-4-1)} [-2\sin 2x - \cos 2x]\right]$$

$$= \frac{1}{2}\left[1 - \frac{1}{15}(-2\sin 2x - \cos 2x)\right]$$

$$P.I = \frac{1}{2}\left[1 + \frac{1}{15}(2\sin 2x + \cos 2x)\right]$$

$$P.I = \frac{1}{2} + \frac{1}{30}[2\sin 2x + \cos 2x]$$

The general solution of Equation (1) is given as

$$y = C.F. + P.I$$

$$y = C_1 e^{-x} + C_2 \cos x + C_3 \sin x + \frac{1}{2} + \frac{1}{30}(2\sin 2x + \cos 2x)$$

3. Solve $(D^2+4)y = \cos 2x$

Soln: Let $(D^2+4)y = \cos 2x$ — (1) be the given LDE with cc.

we have AE, $D^2+4=0$

$(D+2i)(D-2i)=0$

$D = 2i, -2i$

$\therefore$ The root of AE are $2i, -2i$

$\therefore C.F = C_1 \cos 2x + C_2 \sin 2x$

we have $P.I = \frac{1}{D^2+4} \cdot \cos 2x = \frac{x}{2D} \cos 2x$ — (11)

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$$= \frac{x}{2}\frac{1}{D}\cos 2x = \frac{x}{2}\int \cos 2x\,dx = \frac{x}{2}\frac{\sin 2x}{2}$$

$$P.I = \frac{x}{4}\sin 2x \quad -(3)$$

The general solution of Equation (1) is,

$$y = C.F + P.I$$

$$y = C_1\cos 2x + C_2 \sin 2x + \frac{x}{4}\sin 2x$$

4. Solve $(D^2-4)y = \cos^2 x$

Soln: Let $(D^2-4)y = \cos^2 x$ — (1) be the given LDE with cc

we have AE, $D^2-4=0$

$$(D-2)(D+2)=0$$

$$D = (2, -2)$$

∴ The roots of AE are $2, -2$,

$$C.F = C_1 e^{2x} + C_2 e^{-2x} \quad -(2)$$

Consider $P.I = \frac{1}{D^2-4}\cos^2 x$

$$= \frac{1}{D^2-4}\left(\frac{1+\cos 2x}{2}\right)$$

$$= \frac{1}{2}\left[\frac{1}{D^2-4}\cdot 1 + \frac{1}{D^2-4}\cdot\cos 2x\right]$$

$$= \frac{1}{2}\left[\frac{1}{D^2-4}e^{ox} + \frac{1}{D^2-4}\cdot\cos 2x\right]$$

$$= \frac{1}{2}\left[\frac{1}{0-4}\cdot e^{ox} + \frac{1}{-4-4}\cdot\cos 2x\right]$$

$$= \frac{1}{2}\left[\frac{-1}{4}\cdot 1 + \frac{1}{-8}\cdot\cos 2x\right] = \frac{-1}{8}\left[1+\frac{1}{2}\cos 2x\right]$$

$$P.I = \frac{-1}{16}\left[2+\cos 2x\right]$$

The general solution of Equation ① is

$y = C.F + P.I$

$y = C_1e^{2x} + C_2e^{-2x} - \frac{1}{16}[2+\cos 2x]$

5. Solve $(D^2+5D+6)y = \sin^2 x \cos x$

Soln: let $(D^2+5D+6)y = \sin^2 x \cos x$ be the given LDE with CC

we have AE, $D^2+5D+6 = 0$

$(D+2)(D+3) = 0$

$\therefore D = -2, -3$

$\therefore$ The roots of AE are $-2, -3$

$C.F = C_1e^{-2x} + C_2e^{-3x}$

$\sin^2 x \cdot \cos x = \frac{1}{4}(\cos x - \cos 3x)$

Consider $P.I = \frac{1}{4}\left[\frac{1}{D^2+5D+6}\cos x - \frac{1}{D^2+5D+6}\cos 3x\right.$

$\therefore P.I = \frac{\cos x}{20} + \frac{\cos 3x}{12}$

The general solution of Equation ① is

$y = C.F + P.I$

$y = C_1e^{-2x} + C_2e^{-3x} + \frac{\cos x}{20} + \frac{\cos 3x}{12}$

Problems on $X = x^m$

1. Solve $\frac{d^3y}{dx^3} - 2\frac{dy}{dx} + 4y = 3x^2 + 5x - 3$

Soln:- Let $\frac{d^3y}{dx^3} - 2\frac{dy}{dx} + 4y = 3x^2 + 5x - 3$ be the given LDE with CC

$(D^3 - 2D + 4)y = 3x^2 + 5x - 3$ — ①

The Auxiliary Equation is given as,

$(D^3 - 2D + 4) = 0$

$(D+2)(D^2 - 2D + 2) = 0$

$\therefore D = -2, 1 \pm i$

$\therefore$ The roots of AE are $-2, 1+i, 1-i$

$\therefore C.F = C_1e^{-2x} + e^x[C_2 \cos x + C_3 \sin x]$

$$P.I. = \frac{1}{f(D)} \cdot x = \frac{1}{D^3-2D+4}(3x^2+5x-3) = \frac{1}{4}\frac{1}{\left[1-\frac{2D-D^3}{4}\right]}(3x^2+5x-3)$$

$$= \frac{1}{4}\left[1-\left(\frac{2D-D^3}{4}\right)\right]^{-1}(3x^2+5x-3)$$

$$= \frac{1}{4}\left[1+\left(\frac{2D-D^3}{4}\right)+\left(\frac{2D-D^3}{4}\right)^2+\ldots\right](3x^2+5x-3)$$

$$= \frac{1}{4}\left[1+\frac{D}{2}-\frac{D^3}{4}+\frac{D^2}{4}+\ldots\right](3x^2+5x-3)$$

$$= \frac{1}{4}\left[3x^2+5x-3+\frac{1}{2}(6x+5)+\frac{1}{4}(6)\right] = \frac{1}{4}\left[3x^2+5x-3+3x+5/2+3/2\right]$$

$$P.I = \frac{1}{4}(3x^2+8x+1)$$

∴ The general solution of Equation ① is

$$y = C.F + P.I$$

$$y = C_1e^{-2x} + e^x[C_2\cos x + C_3\sin x] + \frac{1}{4}(3x^2+8x+1)$$

2. Solve $(D^3-D^2-6D)y = (1+x^2)$

Soln:- Let $(D^3-D^2-6D)y = (1+x^2)$ be the given LDE with CC.

The Auxiliary equation is given as,

$$D^3-D^2-6D = 0$$

$$D(D-3)(D+2) = 0$$

$$D = 0, 3, -2$$

∴ The roots of AE are $0, 3, -2$

$$C.F = C_1 + C_2e^{3x} + C_3e^{-2x}$$

Consider $P.I = \frac{1}{D(D-3)(D+2)}(1+x^2)$

$$x\frac{1}{(D-3)(D+2)}(1+x^2)$$

$$\frac{1}{(D-3)(D+2)} = \frac{1}{-6}\frac{1}{1-\frac{D^2-D}{6}}$$

$\therefore PI = -\frac{x^3}{18} - \frac{x^2}{6}$

$\therefore$ The general solution is given as

$y = C_1 + C_2 e^{3x} + C_3 e^{-2x} - \frac{x^3}{18} - \frac{x^2}{6}$

3. Solve $(D^2+2D+5)y = 1+x^2$

Soln:- Let $(D^2+2D+5)y = 1+x^2$ be the given LDE with CC.

The Auxiliary equation is equation given as,

$D^2+2D+5 = 0$

$(D^2+2D+1+4) = 0$

$(D+1)^2 = -4$

$D+1 = \pm 2i$

$D = -1 \pm 2i$

The roots of AE are $-1, -2i, -1+2i$

$C.F = e^{-x}[C_1 \cos 2x + C_2 \sin 2x]$

Consider $P.I = \frac{1}{F(D)} \cdot x = \frac{1}{D^2+2D+5}(x^2+1) = \frac{1}{5+2D+D^2}(x^2+1)$

$= \frac{1}{5} \frac{1}{\left[1+\left(\frac{2D+D^2}{5}\right)\right]}(x^2+1) = \frac{1}{5}\left[1+\left(\frac{2D+D^2}{5}\right)\right]^{-1}(x^2+1)$

$= \frac{1}{5}\left[1-\left(\frac{2D+D^2}{5}\right)+\left(\frac{2D+D^2}{5}\right)^2+\ldots\right](x^2+1)$

$= \frac{1}{5}\left[1-\frac{2D}{5}-\frac{D^2}{5}+\frac{4D^2}{25}+\ldots\right](x^2+1) = \frac{1}{5}\left[1-\frac{2D}{5}-\frac{D^2}{25}+\ldots\right](x^2+1)$

$= \frac{1}{5}\left[x^2+1-\frac{2}{5}(2x)-\frac{1}{25}(2)\right]$

$P.I = \frac{1}{5}\left[x^2-\frac{4x}{5}+\frac{23}{25}\right]$ — (3)

$\therefore$ The general solution of equation (1) is given as

$y = C.F + P.I$

$y = e^{-x}[C_1 \cos 2x + C_2 \sin 2x] + \frac{1}{5}\left[x^2-\frac{4}{5}x+\frac{23}{25}\right]$

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4. Solve $(D^3-D^2-6D)y = x^2+1$

Soln:- Let $(D^3-D^2-6D)y = x^2+1$ — ① be the given LDE with CC.

The auxiliary equation is given as,

$$D^3-D^2-6D=0$$
$$D(D-3)(D+2)=0$$
$$\therefore D=0, 3, -2$$

The roots of AE are $0, 3, -2$

$$C.F = C_1+C_2e^{3x}+C_3e^{-2x} \quad — (II)$$

Consider $P.I_3 = \frac{1}{f(D)} \cdot x = \frac{1}{D(D-3)(D+2)} (1+x^2)$

$$\frac{1}{(D-3)(D+2)} = \frac{1}{-6} \frac{1}{1-\frac{D^2-D}{6}}$$

$$P.I = \frac{-x^3}{18} - \frac{x^2}{6} \quad — (III)$$

∴ The General solution of equation ① is given as $y = C.F+P.I$

$$y = C_1+C_2e^{3x}+C_3e^{-2x} - \frac{x^3}{18} - \frac{x^2}{6}$$

Problems on general formulae

1. Solve $(D^2+a^2)y = \sec ax$

Soln: Let $(D^2+a^2)y = \sec ax$ — ① be the given LDE with CC.

we have AE $D^2+a^2=0$

$$\therefore D^2 = -a^2$$
$$\therefore D = \pm ai$$

The roots of AE are $ai, -ai$

$$C.F = C_1 \cos ax + C_2 \sin ax \quad — (II)$$

Consider $P.I = \frac{1}{D^2+a^2} \sec ax$

$$= \frac{1}{(D-ai)(D+ai)} \sec ax$$

$$= \frac{1}{2ai}\left[\frac{1}{(D-ai)} - \frac{1}{D+ai}\right].\sec ax$$

$$P.I = \frac{1}{2ai}\left[\frac{1}{D-ai}\sec ax - \frac{1}{D+ai}\sec ax\right] \quad \text{— (3)}$$

In: Consider, $\frac{1}{D-ai}\sec ax = e^{iax}\int e^{-iax}\sec ax\, dx$

$$= e^{iax}\int \frac{(\cos ax - i\sin ax)}{\cos ax}dx$$

$$= e^{iax}\int (1 - i\tan ax)dx$$

$$= e^{iax}\left[x + \frac{i}{a}\log(\cos ax)\right]$$

$$= e^{iax}\left[x + \frac{i}{a}\log(\cos ax)\right]$$

$$\therefore \frac{1}{D-ai}\sec ax = e^{iax}\left[x + \frac{i}{a}\log(\cos ax)\right]$$

Now we have

$$\frac{1}{D+ai}\sec ax = e^{-iax}\int e^{iax}\sec ax\, dx$$

$$= e^{-iax}\int\left(\frac{\cos ax + i\sin ax}{\cos ax}\right)dx$$

$$= e^{-iax}\int (1 + i\tan ax)dx$$

$$= e^{-iax}\left[1 - \frac{i}{a}\log(\cos ax)\right]$$

$$\frac{1}{D+ai}\sec ax$$

From equation (3),

$$P.I = \frac{1}{2ai}\left[x(e^{iax} - e^{-iax}) + \frac{i}{a}(e^{iax} + e^{-iax})\log(\cos ax)\right]$$

$$\therefore P.I = \frac{x}{a}\sin ax + \frac{1}{a^2}\cos ax \log(\cos ax) \quad \text{— (4)}$$

The general solution of Equation (1) is

$$y = C.F + P.I$$

$$y = C_1\cos ax + C_2\sin ax + \frac{x}{a}\sin ax + \frac{1}{a^2}\cos ax \log(\cos ax)$$

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2. Solve $(D^2-1)y = e^{-x}\sin(e^{-x}) + \cos(e^{-x})$

Soln: Let $(D^2-1)y = e^{-x}\sin(e^{-x}) + \cos(e^{-x})$ be the given LDE with CC.

we have AE, $D^2-1=0$

$\therefore (D-1)(D+1)=0$

$\therefore D = 1, -1$

The roots of AE are $-1, 1$

$\therefore C.F = C_1e^{x} + C_2e^{-x}$

Consider,

$$P.I = \frac{1}{D^2-1}\left[e^{-x}\sin(e^{-x}) + \cos(e^{-x})\right]dx$$

$$= \frac{1}{2}\left[\frac{1}{D-1} - \frac{1}{D+1}\right]\left[e^{-x}\sin(e^{-x}) + \cos(e^{-x})\right]$$

$$P.I = \frac{1}{2}\left[\frac{1}{D-1}\left(e^{-x}\sin(e^{-x}) + \cos(e^{-x})\right) - \left[\frac{1}{D+1}e^{-x}\right.\right.$$

$$\left.\left.\sin(e^{-x}) + \cos(e^{-x})\right]\right] \quad \text{— (3)}$$

we have

$$\frac{1}{D-1}\left[\cos(e^{-x}) + e^{-x}\cdot\sin(e^{-x})\right] = e^{x}\int e^{-x}\left[\cos(e^{-x}) + e^{-x}\cdot\sin(e^{-x})\right]dx$$

Put $e^{-x} = t$

$\therefore -e^{x}\,dx = dt$

$\therefore e^{-x}\,dx = -dt = e^{x}\int(\cos t + t\sin t)(-dt)$

$= -e^{x}\int(\cos t + t\sin t)dt$

$= -e^{x}\left[(\sin t + t(-\cos t) - (1) - (-\sin t)\right]$

$= -e^{x}\left[2\sin t - t\cos t\right]$

Put $t = e^{-x}$

$$\frac{1}{D-1}\left[\cos(e^{-x}) + e^{-x}\cdot\sin(e^{-x})\right] = -e^{x}\left[2\sin(e^{-x}) - e^{-x}\cos(e^{-x})\right]$$

Now we consider

$$\frac{1}{D+1}\left[\cos(e^{-x}) + e^{-x}\cdot\sin(e^{-x})\right]$$

$= e^{-x} \int e^x [\cos(e^{-x}) + e^{-x} \cdot \sin(e^{-x})]\, dx$

$\{ \int e^x [f(x) + f'(x)]\, dx = e^x f(x) \}$

$= e^{-x} [e^x \cos(e^{-x})] = \cos(e^{-x})$

$\frac{1}{D+1} [\cos(e^{-x}) + e^{-x} \cdot \sin(e^{-x})] = \cos(e^{-x})$

From Equation (3) we get

$P.I = \frac{1}{2} [-2e^x \sin(e^{-x}) + \cos(e^{-x})] - \frac{1}{2} \cos(e^{-x}) = -e^x \sin(e^{-x})$

The general solution of Equation (1) is given as

$y = C.F + P.I$

$y = C_1 e^x + C_2 e^{-x} - e^x \sin(e^{-x})$

3. Solve $(D^2 - D - 2)y = 2\log x + \frac{1}{x} + \frac{1}{x^2}$

Soln:- Let $(D^2 - D - 2)y = 2\log x + \frac{1}{x} + \frac{1}{x^2}$ — (1) be the given LDE with cc.

Consider AE, $D^2 - D - 2 = 0$

$(D-2)(D+1) = 0$

$D = 2, -1$

The roots of AE are $2, -1$

$\therefore CF = C_1 e^{2x} + C_2 e^{-x}$ — (2)

we have

$P.I = \frac{1}{D^2 - D - 2} \left[2\log x + \frac{1}{x} + \frac{1}{x^2}\right]$

$= \frac{1}{(D-2)(D+1)} \left[2\log x + \frac{2}{x} - \frac{1}{x} + \frac{1}{x^2}\right]$

$= \frac{1}{(D-2)} e^{-x} \int e^x \left[(2\log x - \frac{1}{x}) + (\frac{2}{x} + \frac{1}{x^2})\right] dx$

$= \frac{1}{(D-2)} e^{-x} \left[e^x (2\log x - \frac{1}{x})\right]$

$P.I = \frac{1}{(D-2)} (2\log x - \frac{1}{x}) = e^{2x} \int e^{-2x} \left[2\log x - \frac{1}{x}\right] dx$

$= e^{2x} \left[\int (2\log x) \cdot (e^{-2x})\, dx - \int e^{-2x} \left[\frac{2}{x} e^{-2x} \frac{1}{x}\right] dx\right.$

$$= e^{2x}\left[2\log x\left(\frac{e^{-2x}}{-2}\right) - \int \frac{e^{-2x}}{-2}\frac{2}{x}\,dx - \int e^{-2x}\frac{1}{x}\,dx\right]$$

$$= e^{2x}\left[-e^{-2x}\log x + \int e^{-2x}\frac{1}{x}\,dx - \int e^{-2x}\frac{1}{x}\,dx\right]$$

$$P.I = -\log x \quad \text{— (3)}$$

The general solution of Equation (1) is given as

$$y = C.F + P.I$$

$$y = C_1 e^{2x} + C_2 e^{-x}\cdot\log x$$

Problems on method of variation of parameters.

1. Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$

Soln:- Let $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$ be the given LDE with cc.

$$(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2} \quad \text{— (1)}$$

we have AE as

$$D^2 - 6D + 9 = 0$$

$$(D-3)^2 = 0$$

$$D = 3, 3$$

The roots of AE are 3, 3

$$\therefore C.F = C_1 e^{3x} + C_2 x e^{3x} = (C_1 + C_2 x)e^{3x} \quad \text{— (2)}$$

we have $y_1 = e^{3x}$, $y_2 = xe^{3x}$

$y_1' = 3e^{3x}$, $y_2' = e^{3x}(3x+1)$

Consider $W = \begin{vmatrix} Y_1 & Y_2' \\ Y_1' & Y_2' \end{vmatrix} = \begin{vmatrix} e^{3x} & xe^{3x} \\ 3e^{3x} & e^{3x}(3x+1) \end{vmatrix}$

$$= e^{6x}(3x+1-3x) = e^{6x}$$

$$\therefore W = e^{6x}$$

By Method of variation of parameters:

$$u = -\int \frac{y_2 x}{w}\,dx = -\int \frac{x e^{3x}}{e^{6x}} \cdot \frac{e^{3x}}{x^2}\,dx = -\int \frac{1}{x}\,dx$$

$u = -\log x$

and $v = -\int \frac{y_1 x}{w}\,dx = \int \frac{e^{3x}}{e^{6x}} \cdot \frac{e^{3x}}{x^2}\,dx = \int \frac{1}{x^2}\,dx = -\frac{1}{x}$ $\therefore v = -\frac{1}{x}$

we have $PI = uy_1 + vy_2$

$PI = -e^{3x}\log x + xe^{3x}\left(-\frac{1}{x}\right)$

$PI = -e^{3x}(\log x + 1)$ —— ③

The general solution of Equation ① is given as

$y = CF + PI$

$\therefore y = (C_1 + C_2 x)e^{3x} - e^{3x}(\log x + 1)$

2. Solve $(D^2 - 2D - 2)y = e^x \tan x$

Soln:- Let $(D^2 - 2D - 2)y = e^x \tan x$ —— ① be the given LDE with cc.

we have AE $D^2 - 2D + 2 = 0$

$(D^2 - 2D + 1) + 1 = 0$

$(D-1)^2 = -1$

$\therefore (D-1)^2 = i^2$

$(D-1) = \pm i$

$\therefore D = 1 \pm i$

The roots of AE are $1+i, 1-i$

$C.F = e^x(C_1 \cos x + C_2 \sin x)$

$C.F = C_1 e^x \cos x + C_2 e^x \sin x$ — ⑪

$y_1 = e^x \cos x$ and $y_2 = e^x \sin x$

$y_1' = e^x[\cos x - \sin x]$

$y_2' = e^x[\sin x + \cos x]$

consider

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x(\cos x - \sin x) & e^x(\sin x + \cos x) \end{vmatrix}$$

$= e^{2x}(\cos x - \sin x + \cos^2 x) - e^{2x}(\cos x \cdot \sin x - \sin^2 x)$

$= e^{2x}[\cos x \cdot \sin x + \cos^2 x - \cos x \cdot \sin x + \sin^2 x]$

$W = e^{2x}[\cos^2 x + \sin^2 x] = e^{2x}$

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we have

$$u = -\int \frac{y_2 x}{W} = -\int \frac{e^x \sin x \cdot e^x \tan x}{e^{2x}} dx$$

$$= -\int \sin x \tan x \, dx = -\int \frac{\sin^2 x}{\cos x} dx$$

$$= -\int \frac{(1-\cos^2 x)}{\cos x} dx = -\int \left(\frac{1}{\cos x} - \cos x\right) dx$$

$$= -\int (\sec x - \cos x) dx$$

$$u = -\log(\sec x + \tan x) + \sin x$$

Now, consider

$$v = \int \frac{y_1 x}{W} dx = \int \frac{e^x \cos x \cdot e^x \tan x}{e^{2x}}$$

$$\int \sin x \, dx = -\cos x$$

$$\therefore v = -\cos x$$

By method of variation of parameter,

$$P.I = uy_1 + vy_2$$

$$= -e^x \cos x \cdot \log(\sec x + \tan x) + e^x \cos x \cdot \sin x - e^x \sin x \cdot \cos x$$

$$P.I = -e^x \cos x \cdot \log(\sec x + \tan x)$$

The general solution of Equation (1) is given as

$$y = C.F + P.I$$

$$y = e^x(C_1 \cos x + C_2 \sin x) - e^x \cos x \log(\sec x + \tan x)$$

3. Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$

Soln:- Let $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$ be the given LDE with CC.

$\therefore (D^2+3D+2)y = e^{e^x}$ — (1)

we have AE $D^2+3D+2=0$

$(D+1)(D+2)=0$

$\therefore D = -1, -2$

The roots are $-1, -2$

$\therefore C.F = C_1e^{-x} + C_2e^{-2x}$ — (2)

we get $y_1 = e^{-x}, y_2 = e^{-2x}$

$y_1 = -e^{-x}$ and $y_2 = -2e^{-2x}$

Consider $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$\begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-3x} \end{vmatrix}$

$= -2e^{-3x} + e^{-3x} = -e^{-3x}$

we have $u = -\int \frac{y_2 x}{W} dx = -\int \frac{e^{-2x} \cdot e^{e^x}}{-e^{-3x}} dx$

$= \int e^x e^{e^x} dx$

$u = e^{e^x}$

now $v = \int \frac{y_1 x}{W} dx = \int \frac{e^{-x} \cdot e^{e^x}}{-e^{-3x}} dx$

$= -\int e^{2x} \cdot e^{e^x} dx = -\int e^x \cdot e^{e^x} e^x dx$

Put $e^x = t$

$\therefore e^x dx = dt = -\int te^t dt = -(t-1)e^t$

$v = -(e^x - 1)e^{e^x} = -e^x e^{e^x} + te^{e^x}$

$v = -e^x \cdot e^{e^x} + te^{e^x}$

by method of variation of Parameter, we get

$P.I = uy_1 + vy_2 = e^{-x} \cdot e^{e^x} + e^{-2x}(e^{e^x} - e^x e^{e^x})$

$P.I = e^{-x} + te^{e^x} + te^{-2x} e^{e^x} - e^{-x} e^{e^x} = e^{-2x} \cdot e^{e^x}$ — (3)

The general solution of Equation (1) is given by

$y = C.F + PI$

$y = C_1e^{-x} + C_2e^{-2x} + e^{-2x} \cdot e^{e^x}$

4. Solve $(D^2-1)y = \frac{2}{1+e^x}$

Soln:- Let $(D^2-1)y = \frac{2}{1+e^x}$ — (1) be the given LDE with CC

we have AE as $(D^2-1)=0$

$(D+1)(D-1)=0$

$D = \pm 1$

∴ The roots of AE are $1, -1$

$C.F = C_1e^x + C_2e^{-x}$ — (2)

we get $y_1 = e^x$, $y_2 = e^{-x}$, $y_1' = e^x$, $y_2' = -e^{-x}$

$$\therefore W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

$= -e^x e^{-x} - e^x \cdot e^{-x}$

$W = -2$

Consider $u = -\int \frac{y_2 x}{W}\,dx = -\int \frac{e^{-x}}{-2}\left(\frac{2}{1+e^x}\right)dx$

$= \int \frac{e^{-x}}{1+e^x}\cdot dx = \int \frac{1}{e^x(1+e^x)}\,dx$

$= \int \left(\frac{1}{e^x} - \frac{1}{1+e^x}\right)dx$

$u = -e^{-x} - \log(e^{-x}+1)$

Now we have $v = \int \frac{y_1 V}{W}\,dx = \int \frac{e^x}{-2}\cdot\left(\frac{2}{1+e^x}\right)dx$

$= -\int \frac{e^{-x}}{1+e^x}\,dx = -\log(1+e^x)$

$v = -\log(1+e^x)$

By method of variation of Parameter, we get,

$P.I = uy_1 + vy_2$

$$P.I = e^{x}\left[-e^{-x} - \log(e^{-x}+1)\right] + e^{-x}\left[-\log(1+e^{x})\right]$$

$$P.I = -1 - e^{x}\log(e^{-x}+1) - e^{-x}\log(1+e^{x})$$

The general solution of Equation ① is given as

$$y = C.F + P.I$$

$$y = C_1 e^{x} + C_2 e^{-x} - 1 - e^{x}\log(e^{-x}+1) - e^{-x}\log(1+e^{x})$$